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A Search for the de Broglie Particle Internal Clock by Means of Electron Channeling

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Abstract The particle internal clock conjectured by de Broglie in 1924 was investigated in a channeling experiment using a beam of ∼80 MeV electrons aligned along the ⟨110⟩ direction of a 1 µm thick silicon crystal. Some of the electrons undergo a rosette motion, in which they interact with a single atomic row. When the electron energy is finely varied, the rate of electron transmission at 0° shows a 8% dip within 0.5% of the resonance energy, 80.874 MeV, for which the frequency of atomic collisions matches the electron's internal clock frequency. A model is presented to show the compatibility of our data with the de Broglie hypothesis.

Keywords Channeling · Silicon · Electrons · Quantum mechanics

In a previous publication [1], we showed data which can be interpreted as a manifestation of the particle internal clock postulated by L. de Broglie in 1924. In the present paper we shall report again this result in the light of a phenomenological calculation that we used as a guide to design the experiment and understand its significance.

At the beginning of quantum mechanics, L. de Broglie [2, 3] associated a particle of mass m_0 in its rest frame with an internal frequency $\nu_0 = m_0c^2/h$ and a wave

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$\psi_0 = a_0 \exp(2\pi i \nu_0 t)$. The wave length $\lambda = h/p$ of a moving particle in the laboratory frame was then deduced by the Lorentz transformation. However, despite the success of the wave hypothesis, the internal clock frequency was never observed, nor is it known if it is an observable or if it ever exists! The aim of the present experiment was to deal with such a possibility, i.e. to put this frequency in evidence for electrons, connecting it with some kind of interaction.

Let us assume that this interaction acts on the electron position in the laboratory with a periodic "amplitude". In the laboratory the electron internal frequency is $\nu = \nu_0/\gamma$ (where γ is the Lorentz factor $(1-\beta^2)^{-1/2}$), and the distance covered during the clock period is

$$L = \beta c/\nu = hp/(m_0 c)^2 \tag{1}$$

A direct and simple observation of this frequency could be to look for a regular pattern of the distances between ionization points of a particle trajectory in an emulsion plate. However, observing a distance of at least 10 μm would require fairly high energy (10^{19} eV protons or 10^{13} eV electrons), that is at the limit of the highest primary cosmic rays. Observing a possible effect with a better control can be done with the help of the axial channelling effect in a thin single crystal [4, 5].

When the incident direction of fast negative projectiles as electrons is close to a major axial direction of a thin single crystal, electrons are attracted towards atomic strings, which they see mainly as continuous positive charges. Then they suffer multiple nuclear scattering effects that are stronger than when the beam is sent along a random direction. As shown in Fig. 1 the probability of transmission through a thin crystal at a small angle to the incident beam direction depends strongly on the crystal orientation. The transmission probability exhibits a large dip centred on the axis direction. However its deepest part appears to be filled by a sharp peak. In other words, observed is a "W" pattern instead of "U". This well known feature is due to the process called "rosette motion": when the incident beam direction is very close to the axial direction, some electrons with a small but non-zero angular momentum with respect to an atomic row may be trapped in a spiral motion around this atomic row. These electrons suffer reduced multiple scattering effects and, thus, have a high transmittivity. The important point is that electrons undergoing rosette motion interact essentially with a single atomic row throughout the crystal. The resonance momentum will be defined as the momentum at which an electron moving parallel to a string of atoms passes atoms with a frequency equal to the de Broglie frequency. At the resonance momentum, the electrons for which the maximum "amplitude" occurs when they pass in front of target atoms will suffer a greater interaction, while those whose maximum "amplitude" falls just in between will travel with a reduced interaction. The mean effect should not be zero because the potential is not linear in z along the atomic string.

We made a phenomenological calculation by a Monte Carlo method in order to choose the optimal experimental conditions. Aside from specifying the experimental set-up, the aim is to reproduce the physical result shown in Fig. 1 and, also, to introduce the internal clock interaction at least approximately. Classical mechanics was used to describe the electron motion. This has been justified specially in our energy range [4]. The main inputs are: parabolic potentials U to describe atomic structures,

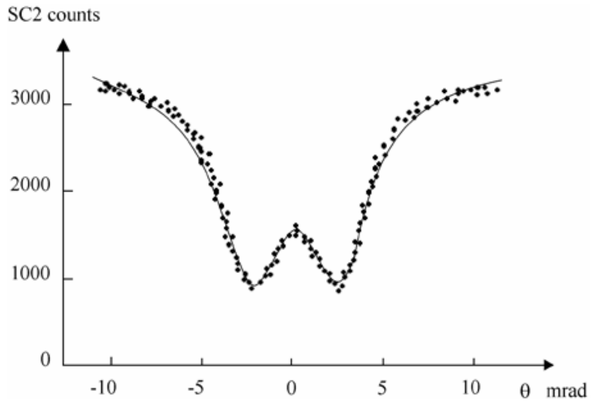

Fig. 1 *Dots*: 80 MeV/c electrons experimental data; counting rate of SC2 vs. crystal tilt angle θ (SC3 $= 5 \times 10^3$). *Full curve*: result of our phenomenological calculation

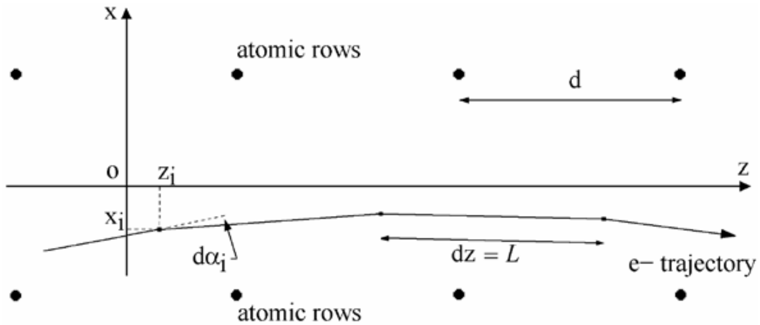

Fig. 2 Electron trajectory between atomic rows as described by our model. $d =$ distance between atomic centers. $d\alpha_i =$ deviation angle at each step. $dz =$ length of one step (y axis is not drawn). A simplified crystal structure is drawn, but in the calculation the more complex Si structure is fully described

a trajectory simulated by a series of segments $dz = L$ (L as defined above), and a classical non relativistic equation of the transverse motion $m\Gamma = U'$ (where m is the total mass $m = m_0\gamma$, Γ the transverse acceleration and U' the first derivative of the potential in the transverse plane). From this equation, the variation angles $d\alpha_i$ in the x direction (Fig. 2) and $d\beta_i$ in the y direction are deduced at each step. The way we proceeded corresponds to an interaction "amplitude" of the internal clock described by a delta function. Since this is not too realistic (it would be better to use a sinusoidal shape), the simplest way was to calculate a fluctuating length at each step with a Gaussian shape $\sigma = L/4$, keeping the non-fluctuating length for the next step. The calculation is done in 3 dimensions, using a separable potential $U(x_i, y_i, z_i) = K x_i^2 y_i^2 z_i^2$ where K is a constant. The deviation in the x direction becomes

$$d\alpha_i = 2LK y_i^2 z_i^2 x_i / m \tag{2}$$

and the deviation in the y direction

$$d\beta_i = 2LK x_i^2 z_i^2 y_i / m \tag{3}$$

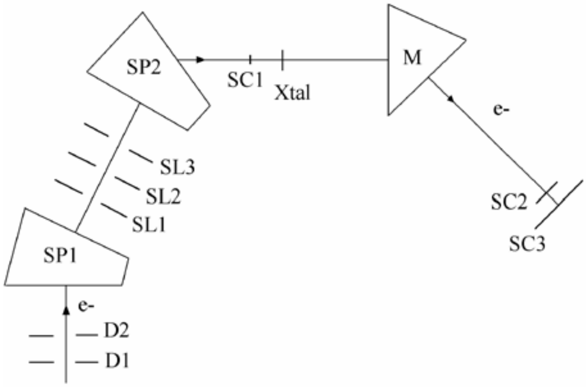

Fig. 3 The experimental set-up. D1 and D2 are collimators (1 mm in diameter, 10 m apart). SP1 and SP2 are two spectrometers (bending magnets and quadrupoles) with SL1, SL2 and SL3 slits. SC1 is a removable scintillator used to center the beam onto the 1 µm thick ⟨110⟩ silicon crystal. M is the last bending magnet. SC2 is a 3×3 mm^2 detector. SC3 is a 10×10 mm^2 monitor

The transmission rate vs. the crystal tilt angle θ (Fig. 1 experimental dots) at 80 MeV/c is well described (Fig. 1 full curve) with only one adjustable parameter K. The normalization is done at 0°. As a function of the electron's momentum, the results show that the event rate at 0° (peak of the rosette motion) is constant except in a sharp momentum range of the order of 1% centred at the resonance momentum. Outside this range the internal clock effect is smoothed and the event rate is the same as the one without the internal clock. There is no free parameter as far as the width is concerned (the parameter σ acts only on the dip amplitude, not on its width). The effect of a longitudinal force induced by the time dependent "amplitude" is found to be negligible. The consequence of this study was that we had to design a set-up able to do very small momentum steps (10^{-3}) without changing the geometrical characteristics of the electron beam.

The experiment was performed during a part of the time allocated to the study of the "planar channelling radiation from 54 to 110 MeV electrons in diamond and silicon" [6] at the Accélérateur Linéaire de Saclay (ALS). The 1-µm thick silicon crystal (lattice constant 5.431 Å) was aligned along the ⟨110⟩ direction (inter-atomic distance along the row $d = 3.84$ Å). The resonance momentum of electrons, for which $L = d$, is 80.874 MeV/c. The experimental set-up is shown in Fig. 3. A detailed description is given in reference [1].

The energy measurements were performed at $\theta = 0°$ with 3 beam momentum adjustments of the spectrometer: 80.874 MeV/c (central), 80.065 MeV/c (-1%) and 81.683 MeV/c ($+1\%$). At each beam adjustment the momentum onto the crystal was varied in small steps (0.083%) by moving the slit SL_2 and the current in M. The total scan domain was $\pm 1.5\%$. The results are presented in Fig. 4. The statistical errors are low ($\sim 0.6\%$) and correspond to the size of the dots.

A 8% dip is seen at 81.1 MeV/c, at a momentum 0.28% higher than the central momentum of 80.874 MeV/c, but this is compatible with the accuracy of our floating wire calibration ($\pm 0.3\%$). The dashed curve shows the result obtained with our model if $dz = L$. Since there is no free parameter in this calculation as far as the width is concerned, the conclusion would be that the observation is badly reproduced. However we can do the same calculation but with $dz = L/2$, i.e. at a frequency twice that introduced by L. de Broglie. It corresponds to the square of our "amplitude" or to the frequency of the "Zitterbewegung" originated by E. Schrödinger [7]. This is shown

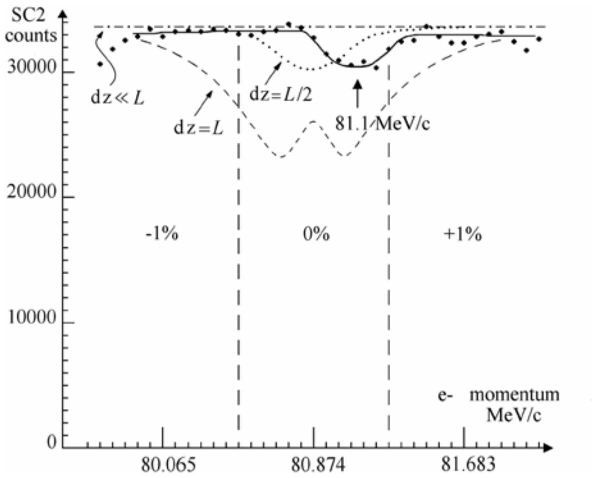

Fig. 4 *Dots*: SC2 experimental counts vs. electron momentum at $\theta = 0°$ for the 3 spectrometer settings, -1%, 0% and $+1\%$, per fixed count in the monitor ($SC3 = 10^5$). The *dots* are corrected data and the *full line* is to guide the eyes. *Dashed curve*: result of our model for $dz = L$. *Dotted curve*: result for $dz = L/2$. *Dashed-doted curve*: result of our model for $dz \ll L$

by the dotted curve. The momentum width (0.5% FWHM) is then correctly reproduced. It would mean that the true resonance momentum is 161.748 MeV/c instead of 80.874 MeV/c, and that we have observed the second harmonic of the de Broglie frequency. Moreover a dashed-doted curve is drawn if $dz \ll L$ in order to show the result of the same calculation in the limit of an infinitesimally small segment.

One must consider which other effects could produce such a resonance. The "Okorokov" effect was already discussed in reference [1]. An alternative explanation would be a frequency associated with pair production, which has the same (or double) resonance momentum of 80.874 MeV/c. It can be estimated using the radiation length X_0 in silicon (~3 cm due to channeling). The thickness of our crystal is 1 µm, which corresponds to $\sim 3 \times 10^{-4} X_0$, i.e. at least two orders of magnitude below our 8% effect. We can conclude that "a frequency associated with a pair production" does not correspond to what we observed.

Finally we cannot explain what we have observed by any known phenomenon. Since the dip width is well fitted by our phenomenological model, it favours a manifestation of the particle internal clock considered by L. de Broglie if the squared amplitude is taken into account or the "Zitterbewegung" originally discussed by E. Schrödinger. A theory implying an observable Zitterbewegung can be found in D.Hestenes's work [8]. Testing again these hypotheses would certainly deserve further experimental and theoretical studies.

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